

SKILL SESSION 17&18

Session 17 :

TASK 1: Implement how to list active jobs, display status information, retrieve job IDs, format output, update state information, and test job listings.

```
[(base) saiabburi@Charishma-MacBook-Air ~ % cd Skill9
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 100 &
[1] 3556
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 200 &
[2] 3559
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 300 &
[3] 3560
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
[1]      running      sleep 100
[2] - running      sleep 200
[3] + running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1]    3556 running      sleep 100
[2] - 3559 running      sleep 200
[3] + 3560 running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs %1
[1]    running      sleep 100
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs %2
[2] - running      sleep 200
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs %3
[3] + running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1]    3556 running      sleep 100
[2] - 3559 running      sleep 200
[3] + 3560 running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % ps -o pid,ppid,stat,command
  PID  PPID  STAT  COMMAND
 3545  3544   S    -zsh
 3556  3545  SN     sleep 100
 3559  3545  SN     sleep 200
 3560  3545  SN     sleep 300
```

```

[(base) saiabburi@Charishma-MacBook-Air Skill9 % fg %1
[1]    running    sleep 100
^Z
zsh: suspended  sleep 100
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
[1]  + suspended  sleep 100
[2]  running     sleep 200
[3]  - running     sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % bg %1
[1]    continued  sleep 100
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
[1]    running    sleep 100
[2]  - running     sleep 200
[3]  + running     sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
[1]    running    sleep 100
[2]  - running     sleep 200
[3]  + running     sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 %
[1]    done        sleep 100
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[2]  - 3559 running    sleep 200
[3]  + 3560 running    sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 100 &
[1] 3564
(base) saiabburi@Charishma-MacBook-Air Skill9 %
[2]    done        sleep 200

```

TASK 2: Design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, and Test Foreground Switching.

```

[(base) saiabburi@Charishma-MacBook-Air ~ % sleep 120 &
[1] 5677
[(base) saiabburi@Charishma-MacBook-Air ~ % jobs -l
[1] + 5677 running      sleep 120
[(base) saiabburi@Charishma-MacBook-Air ~ % fg %1
[1] + running      sleep 120
^C
[(base) saiabburi@Charishma-MacBook-Air ~ % sleep 120
^Z
zsh: suspended  sleep 120
[(base) saiabburi@Charishma-MacBook-Air ~ % jobs -l
[1] + 5678 suspended  sleep 120
[(base) saiabburi@Charishma-MacBook-Air ~ % bg %1
[1] + continued  sleep 120
[(base) saiabburi@Charishma-MacBook-Air ~ % jobs -l
[1] + 5678 running      sleep 120
[(base) saiabburi@Charishma-MacBook-Air ~ % fg %1
[1] + running      sleep 120
^C
[(base) saiabburi@Charishma-MacBook-Air ~ % sleep 10 &
[1] 5698
[(base) saiabburi@Charishma-MacBook-Air ~ % jobs -l
[1] + 5698 running      sleep 10
(base) saiabburi@Charishma-MacBook-Air ~ %
[1] + done      sleep 10
[(base) saiabburi@Charishma-MacBook-Air ~ % wait %1
wait: %1: no such job
[(base) saiabburi@Charishma-MacBook-Air ~ % jobs
(base) saiabburi@Charishma-MacBook-Air ~ % █

```

Session 18 :

TASK 3 : Analyze signal delivery, understand signal masks, review process groups, explore terminal signals, and test signal behavior.

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

int main() {
    sigset_t set;

    sigemptyset(&set);
    sigaddset(&set, SIGINT);

    if (sigprocmask(SIG_BLOCK, &set, NULL) == -1) {
        perror("sigprocmask");
        return 1;
    }

    printf("SIGINT is blocked.\n");
    printf("Press Ctrl+C now.\n");

    sleep(10);

    printf("SIGINT is now unblocked.\n");

    sigprocmask(SIG_UNBLOCK, &set, NULL);

    sleep(5);

    return 0;
}
```

```
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 300 &
[1] 5773
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 5773 running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -STOP 3456
kill: kill 3456 failed: no such process
[(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -STOP 5773
[1] + suspended (signal) sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 5773 suspended (signal) sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -CONT 5773
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 5773 running      sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -TERM 5773
[1] + terminated sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
```

```

[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 300
^Z
zsh: suspended sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 5777 suspended sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % bg %1
[1] + continued sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 5777 running sleep 300
[(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 300 &
[2] 5780

```

```

(base) saiabburi@Charishma-MacBook-Air Skill9 % nano signal_mask.c
(base) saiabburi@Charishma-MacBook-Air Skill9 % gcc signal_mask.c -o signal_mask
(base) saiabburi@Charishma-MacBook-Air Skill9 % ./signal_mask
SIGINT is blocked.
Press Ctrl+C now.
SIGINT is now unblocked.
(base) saiabburi@Charishma-MacBook-Air Skill9 %
(base) saiabburi@Charishma-MacBook-Air Skill9 %
[1] - done sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 %
[2] + done sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % ps -o pid,ppid,pgid,tpgid,stat,command
  PID  PPID  PGID  TPGID  STAT  COMMAND
  5763  5762  5763  5991  S      -zsh
(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 100 &
[1] 6050
(base) saiabburi@Charishma-MacBook-Air Skill9 % ps -o pid,ppid,pgid,tpgid,stat,command -p $!
  PID  PPID  PGID  TPGID  STAT  COMMAND
  6050  5763  6050  6051  SN     sleep 100
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -STOP $!
[1] + suspended (signal) sleep 100
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 6050 suspended (signal) sleep 100
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -CONT $!
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[1] + 6050 running sleep 100
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -TERM $!
[1] + terminated sleep 100
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs

```

TASK 4: Implement how to resume stopped jobs, send continue signals, update job status, maintain the job table, verify execution, and test background recovery.

```
(base) saiabburi@Charishma-MacBook-Air Skill9 % sleep 300
^Z
zsh: suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[[1] - 6514 suspended sleep 300
[2] + 6515 suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % bg %1
[1] - continued sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[[1] - 6514 running sleep 300
[2] + 6515 suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -STOP %1
[1] + suspended (signal) sleep 300

(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[[1] + 6514 suspended (signal) sleep 300
[2] - 6515 suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -CONT %1
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs -l
[[1] + 6514 running sleep 300
[2] - 6515 suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % fg %1
[1] - running sleep 300
^C
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
[2] + suspended sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill -CONT %2
(base) saiabburi@Charishma-MacBook-Air Skill9 % kill %2
[2] + terminated sleep 300
(base) saiabburi@Charishma-MacBook-Air Skill9 % jobs
```